



Machine learning and woody biomasses: Assessing wood chip quality for sustainable energy production

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ARTICLE INFO

Keywords:

Regression

Classification

Predictive modeling

Biofuels

ABSTRACT

This paper explores the integration of Machine Learning techniques in assessing the quality of wood chips, a key biomass source for sustainable energy production. Biomass, specifically wood chips, plays a critical role in transitioning from fossil fuels, which are the primary contributors to global carbon emissions. Traditional methods of evaluating wood chip quality, such as laboratory analysis for moisture content, ash content, nitrogen levels, and heating value, face limitations due to time constraints and variability in material composition. Machine Learning offers a solution by providing real-time, accurate predictions that can optimize combustion efficiency and reduce environmental impact. This study reviews various Machine Learning models like support vector machines, decision trees, artificial neural networks, and partial least squares regression, which have demonstrated high predictive accuracy for parameters like moisture content and heating value. However, challenges remain, particularly in predicting nitrogen and trace elements like chlorine and sulfur, due to biomass heterogeneity. The integration of Machine Learning with remote sensing technologies is proposed as a promising avenue for enhancing real-time quality monitoring throughout the wood chip production chain. Future advancements in model refinement and data acquisition are essential for further optimizing biomass as a renewable energy source.

1. introduction

The reduction of global warming is a significant environmental challenge, with fossil fuel combustion responsible for about 75 % of all CO₂ emissions [1]. Human activities contribute greenhouse gases such as CO₂, CO, CH₄, NO_x, and halocarbons, driving climate change [2]. Global policies aim to mitigate these effects, with a focus on renewable energy incentives. Biomasses, particularly woody materials like wood chips, are key renewable sources due to their abundance and energy density [3]. Eurostat data shows EU wood chip production rising from 88 million m³ in 2004 to 118 million m³ in 2019, followed by a COVID-affected decline to 97 million m³ in 2022 [4]. The EU's biomass capacity also grew from 10 GW in 2000 to 19 GW by 2015, with over 4,000 woodchip-fueled power plants operating by 2017, mainly in Finland, Austria, France, and Germany [5,6].

However, wood chips and other solid biomass fuels have limitations; poor-quality wood can compromise combustion efficiency and low environmental impact. Low-grade wood chips can lead to inefficient combustion, lower energy yields, increased emissions, and unburned

residues, which undermine the intended environmental benefits.

International standards offer methods for assessing wood chip quality, but results can vary due to sampling and analytical process factors [7]. In addition to standard laboratory analyses, innovative methodologies, especially Machine Learning (ML), are increasingly used to monitor in real-time wood chips' quality for enhanced energy efficiency and sustainability [8]. Real-time quality assessments enable power plants to adjust combustion parameters, optimizing energy output, reducing unburned waste, and lowering emissions. Besides boosting combustion efficiency and reducing environmental impact, rapid assessments lower operational costs. That is, continuous monitoring minimizes the need for frequent manual sampling and lab analysis, which are costly and labor-intensive. ML-driven automation streamlines quality control, reduces downtime, and allows plant operators to make quick adjustments for efficient operation.

With this preface, the aim of this paper is to provide a brief description of the qualitative characteristics of wood chips that affect their combustion quality and an overview of new methodologies, particularly ML-based techniques, used to monitor the quality of wood

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<https://doi.org/10.1016/j.biombioe.2024.107527>

Received 16 October 2024; Received in revised form 25 November 2024; Accepted 30 November 2024

Available online 7 December 2024

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chips with innovative, accurate and real-time applications.

2. Wood chips’ main qualitative characteristics

Technical standards ISO 17225-4 [9] and ISO 17225-9 [10] specify the quality characteristics that wood chips must meet to be classified within standard quality classes (Table 1). ISO 17225-4 focuses on wood chips for domestic use, while ISO 17225-9 addresses wood chips intended for industrial use. The main quality characteristics covered by the standards, and which define belonging to one quality class rather than another, include moisture content (MC), ash (AC), nitrogen (N), chlorine (Cl), sulfur (S), heating value (HV) and Particle Size Distribution (PSD).

2.1. Moisture Content

MC is one of the most undesirable but unavoidable components of wood chips. The high thermal inertia of water in wood negatively affects combustion kinetics, reducing the flammability of wood by slowing down the ignition process and lowering combustion efficiency [11]. According to Guerrero et al. [12], when combusted, wood at 25 % MC on wet basis resulted in an 11 % increase in particulate emissions compared to wood at near 0 % MC due to incomplete combustion. Additionally, increasing MC levels has proven to negatively affect combustion efficiency, leading to a higher CO/CO₂ ratio, indicating an inefficient combustion [13]. According to Peng et al. [14], MC can also affect organic compound (OC) emissions. However, the relationship between MC and OC is not always linear. At excessively low MC levels, the combustible skips the dehydration stage and enters directly into the devolatilization process [14], leading to a shortage of oxygen in the combustion chamber and increasing CO emissions. Moreover, MC and inadequate ventilation of wood chip piles during storage favors fungal growth [15], posing a health risk to humans, and accelerates biological oxidation, which can result in pile’s self-ignition [16,17]. It is therefore clear why it is crucial to ensure the monitoring of MC at all stages of the supply chain, particularly during storage and feeding the fuel into the combustion chambers in power plants.

2.2. Ash Content

When referring to the AC of wood chips, it signifies the inorganic fraction, primarily composed of the bark and other inert materials present within the wood. This fraction does not participate in the oxidation of the carbon-based components during combustion and remains as residue after the process is complete [18]. AC, similarly, to MC, negatively impacts thermal efficiency, as the inorganic fraction of the total fuel mass increases, the organic fraction available for combustion decreases [19,20]. Moreover, according to Sun et al. [21], higher AC slows ignition propagation, resulting in reduced combustion efficiency.

AC can be categorized into fly AC and coarse AC. The former is carried away in the chimney with flue gases, whereas the latter settles on the combustion bed. So, while fly AC is an environmental issue as it is dispersed into the atmosphere, coarse AC is an obstacle for thermal power plants as it is an inert residue which leads to disposal costs and, due to its chemical composition, to corrosion of systems. The composition of AC varies with the type of wood being burned. Wood chips with more bark produce AC rich in elements such as Cl, sodium (Na), aluminum (Al), and silicon (Si). In contrast, bark-less wood generally produces more fly than coarse AC and contains higher concentrations of potassium (K) and heavy metals [22]. While lower quality wood chips could still be used, requiring however pre-treatments such as hydro-thermal carbonization to reduce AC [21], the use of high-quality wood chips is crucial to ensure power plant’s thermal efficiency [23].

2.3. Nitrogen

Wood is a complex biological material consisting mainly of carbon (C), hydrogen (H), oxygen (O), N, and other trace elements [24]. These elements combine to form organic compounds, lignin, cellulose, and hemicellulose, that constitute the structural framework of wood.

N is present in trace amounts, usually less than 1 % on dry basis [25] and may vary depending on the tree’s age at the time of felling [26] and on the environmental characteristics of the tree’s growth area [27,28]. N is an unfavorable element because inevitably increases the emission of NO_x during combustion [29–31]. Technical standards do not take C and H into account when defining quality classes, but only N, because of its negative effect on the overall fuel quality.

2.4. Heating Value

The HV is a key indicator of the energy released during combustion, making it an essential measure of a biofuel’s efficiency [32]. The chemical composition of wood, which includes C, H, O and N, along with non-combustible inorganic materials like AC, plays a crucial role in determining its HV [33]. HV is also affected by MC, which negatively impacts combustion’s kinetics [34]. It is important to consider that some of the heat from condensation, the latent heat recovered from water vapor in exhaust gases, can be captured and reused, enhancing the overall energy efficiency of combustion systems. The Higher HV (HHV) indicates the total energy released during combustion, including the heat from condensing water vapor. In contrast, the Lower HV (LHV) reflects the energy available, excluding the heat of vaporization.

Therefore, accurately determining the HV is essential for evaluating the sustainability and cost-effectiveness of using wood chips for power generation [35].

Table 1
Excerpt of wood chips’ quality parameters, regulations describing how they are measured, main instrumentation required according to the regulations and a brief description of the procedure.

Parameters	Technical Standards	Instrument	Methodology
Moisture	ISO 18134-3:2023	Thermo-Ventilated Oven	The sample is oven-dried at 105 °C, and moisture percentage is calculated from mass loss.
Ash	ISO 18122:2022	Muffle or Thermogravimetric Analyzer	Ash content is measured by weighing the residue after heating the sample to 550 ± 10 °C, possibly using automatic equipment.
Nitrogen	ISO 16948:2015	Elemental Analyzer	A sample is burned in oxygen, producing gases. Nitrogen oxides are reduced, impurities removed, and nitrogen, carbon dioxide, and water vapor are analyzed using gas analyzer instruments.
(Higher) Heating Value	ISO 18125:2017	Calorimeter	The sample is ignited in an oxygen-filled bomb in a water chamber, generating heat that raises the water temperature, which is measured to calculate the sample’s energy content.
Chlorine & Sulfur	ISO 16994:2016	Ion Chromatograph	Total Cl and S are determined by burning the sample in an oxygen bomb and collecting the resulting acidic gases in a solution, followed by analysis using ion chromatography.
Particle Size Distribution	ISO 17827-1:2016	Sieving Machine	The material is sieved by rotary movements a series of sieves with decreasing mesh sizes. This process sorts particles by size, allowing assessment of the particle size distribution.

2.5. Chlorine and Sulfur

Wood chips generally contains very low concentrations of Cl and S, approximately 0.01 % on dry basis for the former and 0.01–0.03 % on dry basis for the latter. However, these elements, which are naturally present in biomass, are of major concern [36]. During combustion, Cl and S react with alkali, especially potassium (K) leading to the formation of KCl and K₂SO₄ [37], corrosive agents that damage the surfaces of biomass boilers and combustion chambers [38]. Additionally, both S and Cl influence the thermal properties of ashes, leading to their melting at lower temperatures [39], constituting an additional negative impact for power plants [40]. Lastly, during the combustion, the oxidation of S leads to the production of sulfur dioxide (SO₂), one of the main contributors to air pollution [41].

2.6. Particle Size Distribution

Particle size distribution (PSD) in wood chips is critical for handling, storage, and combustion, affecting both efficiency and operational consistency [42]. It influences airflow and drying during storage, while irregular sizes may clog automated systems and impact machine productivity. Additionally, particle size affects combustion efficiency and emissions: finer particles burn faster but often produce more emissions. Proper classification ensures consistent energy output and efficient processing.

3. Introduction to Machine Learning's models

Originating in the 1950s [43,44], ML is a multi-disciplinary field, branch of artificial intelligence, that involves the development of statistical models that can learn from data [45,46]. The models used in ML can be grouped according to their operating principles in Supervised Learning (SL) and Unsupervised Learning (UL) models [47,48].

SL utilizes part of the data as a training dataset to teach models to produce desired outputs [49]. The training dataset contains pairs of inputs and their correct corresponding outputs, allowing the model to learn. The performance of the model is then assessed by a loss function, which allows then to iteratively reduce the model's error through optimization processes [50]. To ensure the model generalizes well to unseen data, cross-validation can also be employed. Cross-validation partitions the data into subsets, trains the model on some, and tests it on others. It averages the results to give a better estimate of model performance and prevent overfitting.

SL address classification and regression problems. While classification algorithms predict (or classify) discrete values, regression algorithms are used to predict continuous values [51].

In contrast to SL models, UL models allow the partitioning of unlabeled datasets into subgroups [52,53]. UL makes it possible to identify hidden patterns and properties of the analyzed dataset, dividing the data into different clusters [54]. UL also incorporates data reduction techniques, which focus on decreasing the dimensionality of data or compressing its representation by selecting and preserving the most significant features or patterns [55].

A brief explanation of the operation and logic of the most common models will be given in the following sections.

3.1. Regression methods and Partial Least Square

Linear regression is a fundamental statistical technique used to model the relationship between a continuous dependent variable and one independent variables [56,57]. The relationship between dependent and independent variables can be summed up by the following formula (1) [58]:

$$Y = \beta_0 + \beta_1 X \quad (1)$$

Where Y is the dependent variable, thus the predicted one, X is the independent variable, β_0 is the intercept of the function, thus the value of Y when $X = 0$, and β_1 is the weight, or slope, of the value X . The intercept and the slope are also known as coefficients. Whenever linear regression involves more than one independent variable, the equation extends to (2), accounting for each additional factor [58]:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_n X_n \quad (2)$$

Where Y is the dependent variable, β_0 is the intercept, and $\beta_1, \beta_2, \dots, \beta_n$ are the weights for each independent variables $X_1, X_2, \dots, X_n$.

One of the most common regression models is the Ordinary Least Square Regression (OLS) [59]. OLS estimates the coefficients of the linear regression equation by minimizing the differences between the observed values and the values predicted by the model [60].

When the relation between Y and X cannot be assumed to be linear, Non-Linear regression (NR) methods are used [61]. An example of a Non-Linear function is the Polynomial Regression [62] which, as seen in function (3) [58], enhances the model's flexibility by including exponentiation of the original variables as additional predictors, allowing it to more effectively capture and fit non-linear relationships [58,63].

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + \beta_3 X_1^3 + \dots + \beta_n X_n^n \quad (3)$$

In high-dimensional data contexts, traditional linear and non-linear regression methods may not be applicable. Partial Least Squares (PLS) provides a robust alternative. PLS is a multivariate statistical method for data with high multicollinearity among independent variables [64]. It works by creating new variables, known as latent variables, from the original independent variables. Latent variables are designed to capture the most significant relationships between the dependent and independent variables [65,66]. Thus, PLS reduces the complexity of the data while maximizing the correlation between the predictors and the response variable. Once the latent variables have been extracted, PLS performs regression using the latent variables instead of the original predictors [67,68]. Lastly, PLS-Discriminant Analysis (PLS-DA) is a variant of PLS specifically designed to classify categorical data. It assigns threshold values to the latent variables, enabling the assignment of class labels to new data points based on their classification [69].

Independently of the model utilized, accuracy metrics are used to evaluate prediction performance. Among the most commonly used are R-squared (R^2) and Root Mean Square Error (RMSE). R^2 indicates the proportion of variance in the dependent variable explained by the model, with values closer to 1 signifying a better fit. RMSE measures the average prediction error in the same units as the dependent variable, providing insight into model accuracy. Together, these metrics assess how well a regression model explains data variability and makes accurate predictions.

3.2. Support Vector Machine

Proposed by Vapnik et al. [70] in the mid-1990s, Support Vector Machine (SVM) is a SL algorithm used for both classification (Support Vector Classifier, SVC) and regression (Support Vector Regression, SVR) problems (Fig. 1a & b).

SVC is applied to both binary and multi-class classification tasks. It works by identifying the optimal boundary separating different classes [71,72]. In binary classification, this boundary is represented by a line. When the dataset is not linearly separable, a kernel function allows for the transformation of the dataset into a higher-dimensional space, which aims to maximize the distances between data points (Fig. 1c). This transformation allows the identification of a plane, or hyperplane, that can linearly separate the data into different classes [73–75]. Given that multiple hyperplanes can separate different classes, the algorithm selects the optimal decision boundary by maximizing the distance, or margin, between points of different classes while minimizing classification errors [70,72]. The margin can be visualized as lines, called

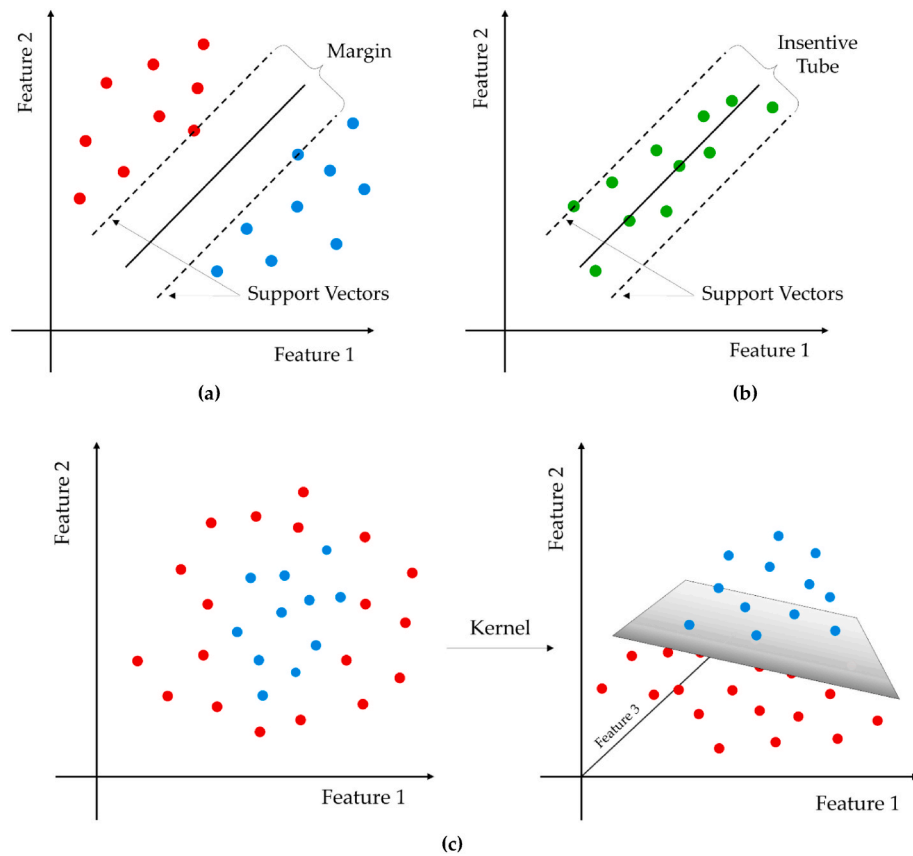


Fig. 1. (a) Binary classification of a bivariate dataset performed via SVC. This line is positioned in such a way that it maximizes the margin between the nearest data points of each class; (b) Graphical representation of a regression function obtained via SVR. The dotted lines represent the tolerated error allowed during prediction; (c) Blue and red dots represent the classified samples. (c) Graphical representation of how a non-linearly separable dataset is transformed via a kernel function and then separated via a hyperplane. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Support Vectors, parallel to the hyperplane that intersect the points closest to it [76].

SVR, on the other hand, aims to identify a regression function that approximates the majority of the data within a specified margin of tolerance defined by a value ϵ , known as the ϵ -insensitive margin [77, 78]. The objective is to find a function that deviates from the actual target values by no more than ϵ for most of the data points [77–79]. SVR also employs kernel functions to handle non-linearity by mapping the input space into a higher-dimensional space where linear regression techniques can be applied to find the best-fit function [80].

3.3. Decision Trees

A Decision Tree (DT) is a non-parametric SL algorithm used for both binary classification (Classification DT, CDT) and regression (Regression DT, RDT) problems [81,82]. DTs can be visualized as flowchart-like structures that begin with a root node (Fig. 2). The root node represents the initial decision point, which then splits into branches, each representing possible outcomes or conditions. These branches lead to intermediate nodes, which continue to split based on further decisions, eventually reaching leaf nodes, thus the final outcomes. Each non-leaf

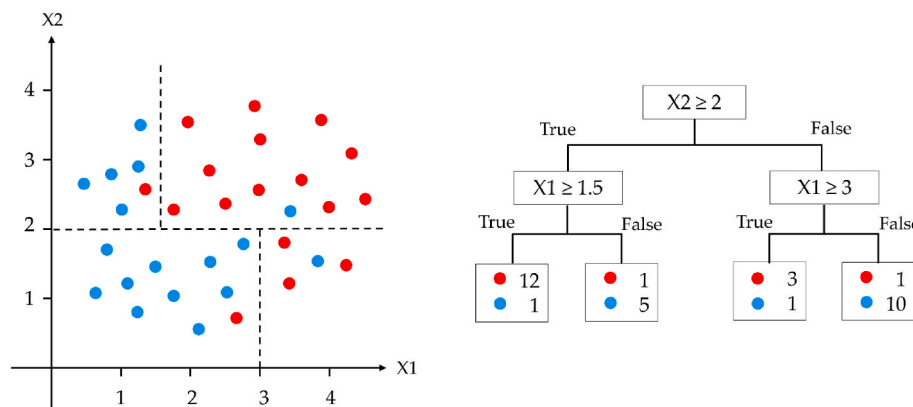


Fig. 2. – Illustration of a CDT. The root node initiates the first data split based on the feature and threshold that provide the best separation. The decision tree then continues splitting at subsequent nodes, refining the classification with each decision. In this case, the process is constrained by a stopping criterion, which limits the depth of the tree to two levels, meaning the tree performs only two splits before reaching its classification decisions.

node contains a split rule or decision function that assesses a specific feature of the input data to divide it into distinct branches [83,84]. Node splitting stops when the leaf nodes achieve complete purity (i.e., they contain only data points of a single class) or when other stopping criteria, such as maximum tree depth or minimum number of samples, are met [85,86].

Specifically, CDT divides the data into different categories or classes based on the features. Starting at the root, the tree selects the feature and threshold that best separates the data into classes. This process is repeated recursively for each subset of the data, creating branches and nodes until it reaches the leaves. Each leaf represents a class label, which is the most common class among the data points that fall into that leaf [87–89].

RDT selects an independent variable and a threshold to divide the data into groups. This process continues recursively, dividing the data until it reaches the leaves. For each leaf, a prediction is made for the dependent variable, usually by calculating the average value of the dependent variable for the data in that leaf [87–89].

To evaluate whether the dataset is properly split, DT typically uses impurity metrics, such as Entropy or the Gini index for CDT and Sum Squared Error (SSE) for RDT. For example, the Gini index measures the probability of misclassification for a randomly chosen element from the dataset, based on the class distribution in the node [90,91], while the SSE is the sum of squared differences between each data point and the mean of the node [89,92].

3.4. Random Forest

Originally proposed in the 1995 by Tin Kam Ho [93] and subsequently refined by Breiman [94] in the early 2000s, Random Forest (RF) is an ensemble learning method for both classification and regression tasks. The term "ensemble learning" refers to the approach of combining multiple algorithms to improve overall performance. In RF's case, it employs multiple decision trees to enhance its predictive accuracy [93, 95,96]. Moreover, RF includes other methods such as bagging [97,98], which is used when modelling DTs. In this process, RF generates new datasets by randomly selecting samples from the original dataset, allowing for the possibility of re-selection, until the new dataset matches the size of the original [99,100]. DTs are then constructed using these modified datasets, with an important distinction: at each node, the algorithm randomly selects a subset of features and identifies the optimal split within that subset [95–97]. This approach ensures that each DT in the RF is based on a distinct dataset and utilizes different features, leading to a diverse ensemble of trees.

Lastly, RF predictions are made by combining the outputs of multiple DT. For classification problems, each tree casts a vote, and the majority

vote decides the outcome, while for regression tasks, the average of the trees' predictions is taken [101,102]. This approach reduces overfitting and improves accuracy by leveraging the diversity of the trees.

3.5. Artificial Neural Network

Artificial Neural Networks (ANN) are complex models with many variants and capabilities. An ANN is a mathematical model that emulates the information processing of biological neural networks (Fig. 3). It is structured in layers of interconnected nodes, or neurons, with each node executing a specific task [103,104]. Specifically, nodes are mathematical functions, which take data as input, process it and produce an output [105]. They can be used for both SL and UL tasks [106], this section however, focuses on the structure of a basic ANN for SL applications.

The simplest ANN's structure is composed of three types of layers: the input layer, which receives the initial data; the hidden layer, where the computations are performed; and the output layer, which produces the final result [103,107].

The input layer handles the raw data, consisting of numerical values representing various features. Each node in this layer corresponds to a specific variable and does not perform computations [103,104].

In the hidden layers, neurons perform computations by calculating a weighted sum of their inputs from the previous layer (which, in a simple ANN, is the input layer), adding a bias term, and applying an activation function to produce their output [103,105]. Activation functions add non-linearity to the data [108]. A widely used activation function is the sigmoid function that scales data into a range between 0 and 1. Without activation functions, ANNs would only perform a series of linear transformations on the input, limiting its capacity to handle complex patterns in multivariate dataset [109]. Only the output values that exceed a specific threshold are passed to the next hidden layer or the output layer.

The output layer provides the final prediction or classification based on the processed data. Neurons in this layer compute a weighted sum of inputs from the last hidden layer, add once more a bias, and apply an activation function [110]. Thus, the activation function is applied in both the hidden and output layers, but while in the hidden layers it is crucial for learning complex patterns, in the output layer it transforms the final output to match the problem requirements, such as converting the output to values between 0 and 1 for binary classification problems [111].

In SL, the strength of ANN comes from their ability to learn from examples through a training process. During training, the network adjusts its weights based on the difference between its actual output and the desired output. This adjustment is carried out using the back-propagation algorithm [112]. Backpropagation works by first

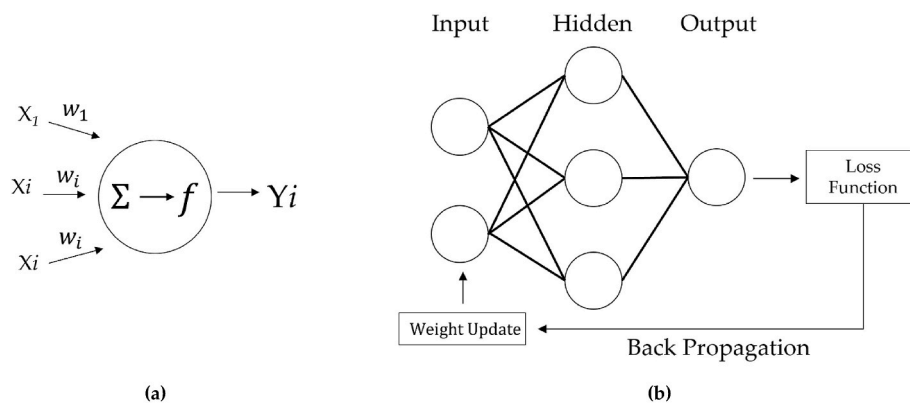


Fig. 3. – (a) Structure of ANN's fundamental unit: a neuron. Each neuron takes multiple inputs (X_i), applies a weight (W_i) to each, sums them up and then passes the result through an activation function, generating an output (Y_i). (b) Structure of a simple ANN, consisting of interconnected layers: an input layer, hidden layers, and an output layer, with each neuron in one layer connected to neurons in the next. During backpropagation, errors from the output layer are propagated backward through the network, calculating gradients for each weight. The weights are updated using these gradients to reduce the network's prediction error.

calculating the error between the network’s prediction and the actual result. It then propagates this error backward through the network, updating each weight to minimize the overall discrepancy. By iteratively refining these weights, the network gradually improves its accuracy and performance over time [113].

4. Machine Learning Applied to wood chips’ quality assessment

Combining traditional laboratory analyses with faster, non-destructive methods is a powerful approach in developing predictive ML models. Laboratory analyses provide precise and detailed measurements when following standard procedures, while non-destructive data collection methods offer quicker and less invasive insights. Integrating data from both sources enables the creation of models that harness the strengths of each approach, allowing for faster and more efficient assessments without compromising accuracy. Non-destructive data often serve as predictors, estimating variables that would otherwise require time-consuming or costly laboratory analyses.

The following sections review the scientific literature on the use of ML models as an alternative to standard analytical methods for assessing wood chip quality. Table 2 provides a summary of the main algorithms used to predict specific parameters.

The references cited in the following chapters were sourced from the Scopus and Web of Science databases. The search included the following keywords: “wood chip AND machine learning”, “biofuel AND machine learning”, “wood AND machine learning”, “woody biofuel AND machine learning”, “wood chip AND machine learning AND moisture content”, “wood chip AND machine learning AND heating value”, “wood chip AND machine learning AND ash content”, “wood chip AND machine learning AND nitrogen content” and “wood chip AND machine learning AND particle size”. During the literature research, which spanned from May to September 2024, no time filters were applied.

To the best of the authors’ knowledge, no ML models have been developed to predict PSD, neither CI and S directly on wood chips prior to combustion. For this reason, no specific sections are presented for these parameters.

4.1. ML and Moisture Content

Due to its high importance in the context of energy efficiency and biological stability of wood chips during storage, MC is one of the most commonly studied parameters when it comes to ML modelling. Literature includes studies that propose both MC prediction and classification models, using data collected from both laboratory analyses and a variety of different sources.

For instance, Daassi-Gnaba et al. [114] developed a SVC model to

classify wood chip piles into different MC classes via a “one vs one” multiclass separation. The model was trained using wood chip permittivity data, collected via a radio frequency electromagnetic module, along with real MC measurements obtained through a 105 °C drying oven, in accordance with standard methodology [115]. The SVC model achieved a 95 % accuracy, suggesting the effectiveness of using SVC combined with electromagnetic principles to accurately determine the MC of piles of woodchips.

In a further study, Daassi-Gnaba et al. [116] focused on regression rather than classification, presenting a second radio frequency-based model, employing kernelized SVR. The model performed effectively, achieving a prediction bias of about 1.8 % compared to the actual MC. Moreover, the model showed higher accuracy for softwood chips than for hardwood chips. However, the significant decrease in analytical time compared to traditional methods largely compensates for the effect of bias.

Suthar et al. [117] classified wood chips based on their MC using models built from data obtained through an innovative approach, namely the Channel State Information (CSI) from Industrial Internet of Things (IIoT) Wi-Fi systems. CSI operates by measuring how Wi-Fi signals interact with a sample, capturing changes in signal amplitude and phase that reflect the sample’s properties, such as MC [118]. The processed CSI data, combined with MC data collected through traditional standard methods, has been used to develop both SVC and ANN models, achieving high classification accuracy of approximately 95–96 %. However, the authors noted that the high accuracy was based on controlled ‘laboratory-scale’ tests, highlighting the need for further research to assess the model’s effectiveness and feasibility for larger-scale, real-world applications.

In their study, Wu et al. [119] proposed a novel method for detecting the moisture content of wood components using Wi-Fi Channel State Information (CSI) combined with a Bimodal Deep Extreme Learning Machine (BDELM). Given that moisture content plays a crucial role in the structural health of wood, and existing detection methods tend to be costly, time-consuming, or sensitive to environmental factors, this research introduces a more efficient, non-destructive approach. The system utilizes commercial Wi-Fi devices to gather CSI data, which is processed using the BDELM model to accurately detect moisture levels in various wood components. The results demonstrated a classification accuracy exceeding 96 %, highlighting the potential of this method for practical applications in monitoring the health of wooden structures.

Ahmed et al. [120] also evaluated the accuracy of SVR and ANN, and PLS models for MC prediction in wood chips, when using near infrared (NIR) data. All models led to high accuracies with a RMSE of 1.96 %, 2.01 % and 2.31 % respectively, and R² of 0.98 each. These results suggested that NIR technology, paired with data pre-processing, could be a valid alternative to traditional MC assessment.

Amaral et al. [121], further investigated the applicability of NIR paired with ML, by developing a regression PLS model and a classification PLS-DA model. The study focused on was the assessment of MC in purposely conditioned wood chips at different MC. Both models demonstrated great potential, though they exhibited higher error rates for wood chips with greater MCs. For instance, the PLS-DA model correctly classified 98 % of samples with an initial MC on dry basis between 0 and 40 %, but its accuracy dropped to 77 % for samples for higher MC classes. Similarly, in the PLS model, the RMSE of external validation increased from 2 % for lower MC to 12.5 % for higher MC, indicating reduced predictive accuracy as moisture levels rose.

Despite the promising results, both Ahmed et al. [120] and Amaral et al. [121] highlighted that the need for extensive data pre-processing, and the selection of different wavelengths between models, could lead to high computational costs, thus making real-time or large-scale applications challenging.

Lastly, Rahman et al. [122] developed two innovative image-based ANN models to predict the MC of wood chips, obtained in two different power plants, using RGB images. The models were trained on 1,

Table 2
Each row lists an algorithm, its associated predicted parameter, and the citation number from the bibliography included in paragraph 4.

Algorithm	Parameter	Bibliography
Linear or Non-Linear Regression	Ash Content	125, 133, 137, 139
	Heating Value	
SVM	Moisture	114, 116, 117, 120, 138, 139
	Content	
DT	Heating Value	133
	Ash Content	
RF	Ash Content	126, 133
	Heating Value	
ANN	Moisture	117, 119, 120, 122, 139
	Content	
PLS/PLS-DA	Heating Value	120, 121, 129, 130, 131, 132, 135, 136, 140
	Moisture	
	Content	
	Ash Content	
	Nitrogen	
	Heating Value	

600 RGB, categorized into dry, medium, and wet MC classes. Ground truth labels generated through the traditional oven-drying method. The two ANNs varied in architectural complexity, leading to high classification accuracies between 87 % and 91 %. However, the study faced some limitations, such as the time-consuming data collection process and significant computational costs.

In conclusion, the development of ML models for wood chip MC prediction has shown great promise in improving efficiency and accuracy, especially with models such as SVC, SVR, ANN, and PLS. However, challenges remain in balancing computational costs, data collection time, and scalability for real-world industrial applications.

4.2. ML and Ash Content

Due to the heterogeneous nature of wood and its variable inorganic chemical composition, the prediction of the AC of wood chips can be a challenge [123,124]. However, more and more researchers are looking for alternative and rapid methods to determine the AC of wood chips that can replace the standard methods, due to the importance of this parameter.

In this context, Nosek et al. [125], provided a basis for predicting AC by investigating the relationship between the bark content of wood chips and AC. The authors applied OLS regression to successfully model this relationship, achieving a R^2 of 0.99. The model exhibited a high degree of reliability in predicting AC based on bark proportion, thus establishing a foundation for future ML approaches to investigate more intricate interactions between AC and other wood chip parameters.

In a compelling study, Morris et al. [126] used a five-year dataset obtained through laboratory analysis, to examine the relationships among 30 qualitative parameters, including chemical, physical, and energetic features, as well as AC. The authors developed a RF regression model, achieving a strong fit on the training data with R^2 of 0.97 and RMSE of 0.29 %. However, when applied to the test data, R^2 dropped to 0.78 and the RMSE increased to 0.74 %, suggesting possible overfitting and a difficulty in generalizing unseen data. The accuracy's decrease was attributed to the high heterogeneity of the fuel data. Moreover, the contrasting results with other studies, where RF models demonstrated high accuracy in predicting various parameters [127,128], indicate that predicting AC poses a distinct challenge.

Given the widespread application of NIR spectroscopy for wood chips' MC assessment, it has also been evaluated as a tool for assessing wood chips' AC. For instance, Mancini et al. [129] explored the application of NIR spectroscopy, coupled with laboratory data as 'ground truth', to assess the AC of wood chips, focusing on the potential impact of wood chips' scattering effects. The study applied Extended Multiplicative Scatter Correction (EMSC) factors in combination with principal component analysis to develop a corrected PLS model (CORR-PLS). When predicting AC, the model outperformed traditional PLS models, reducing the RMSE from 0.46 % to 0.44 % and increasing the R^2 from 0.78 to 0.80. However, incorporating EMSC factors increased the model's complexity, making its application and interpretation more challenging and limiting its broader use. In a further study, Mancini et al. [130] applied NIR spectroscopy and PLS in order to model the relationship between AC and HHV. When predicting AC, the cross validated PLS model achieved promising accuracies, with a RMSE of 0.47 % and a R^2 of 0.77. However, the model was developed using a relatively small sample size, which may not fully capture the broad diversity of wood chip types that could be utilized by thermal power plants. Nonetheless, the study showed that the basis for NIR-PLS modelling of AC is promising and could be effectively developed in future research.

Similarly, Pitak et al. [131] explored a method to predict the quality of commercial wood pellets using NIR hyperspectral imaging coupled with PLS algorithms. The researchers focused on predicting key properties such as AC, fuel ratio (FR), volatile matter (VM) and fixed carbon (FC) by analyzing the pellets with a hyperspectral imaging system.

Various spectral preprocessing techniques and wavelength selection algorithms, including interval successive projections algorithm (iSPA) and interval genetic algorithm (iGA), were applied to improve model performance. Their results showed that standard normal variate spectra with iSPA-100 wavelengths were optimal for predicting VM, FC, and AC. The models demonstrated R^2 of 0.75, 0.81, 0.82, and 0.87 for FR, VM, FC, and AC, respectively. Additionally, color distribution mapping was applied to visualize and monitor pellet quality during the thermal conversion process, enabling real-time quality control in industrial applications.

Another application of NIR for the prediction of the AC has been presented by Loureiro et al. [132]. In this study, the authors used NIR, laboratory data and a cross validated PLS model for predicting AC on wood of fast-growing species, with promising results. The trained model performed well, achieving an RMSE of 0.06 % and an R^2 of 0.96. However, when tested on unseen data, the RMSE increased to 0.10 %, and the R^2 dropped to 0.93. Despite its accuracy, the model is highly specialized for fast-growing species, which restricts its broader applicability.

Lastly, Plankenbühler et al. [133], developed a system that uses image analysis and regression models to indirectly predict AC. Specifically, the system predicts the mixture ratios of wood chips and forest residues by capturing images of the fuel mixture, extracting features such as texture and brightness, and applying methods like OLS, DT, and RF. AC is then calculated using linear interpolation from the known AC values of pure wood chips and forest residues. This approach enables real-time monitoring and control of fuel quality, enhancing the efficiency of biomass plants through automated fuel assessment. The system achieved high predictive accuracy, with R^2 values consistently exceeding 0.87 for mixture ratio predictions. However, the system had certain limitations, including its focus on only two types of biomass and the need for high-quality images to ensure accuracy. Moreover, the prediction of AC relied on a simplified linear interpolation model, which may not fully account for the complexity of biomass mixtures.

In conclusion, while various innovative methods show promise for predicting AC in wood chips, the challenge remains due to the material's inherent heterogeneity. Future advancements in ML and other frequently used technologies could further improve accuracy and scalability across diverse biomass types.

4.3. ML and Nitrogen

Predicting N content is particularly challenging due to its inherent variability, which is strongly influenced by factors like soil quality, nutrient levels, and tree age [26,134]. N content in wood is generally low and often overshadowed by dominant components like C and MC, making it harder for ML models to detect patterns. This challenge is further compounded by small or unbalanced datasets, which can lead to overfitting and poor generalization in N predictions. Nevertheless, some researchers have explored the possibility of predicting N content in different types of biomass, though not specifically in wood chips. A few examples are provided below.

Li Yanjie et al. [135] focused on predicting N content in slash pine (*Pinus elliottii*) tissues, such as needles, trunk, branches, and roots, pairing NIR spectroscopy, laboratory data and PLS. NIR was effective, especially in branches and needles, where models achieved strong predictability with R^2 values up to 0.77 and RMSE of approximately 0.78 %, while root models were less accurate, with R^2 around 0.42 and RMSE of 0.82 %. The analysis identified key spectral regions linked to N content, showing NIR's potential as a reliable tool for the rapid assessment of N content. However, the models faced limitations, including reduced accuracy when applied to roots, poor transferability between different tissue types, and the lack of an external validation set from a different year or environment. Additionally, possible interference from other compounds like phenolics might have introduced some bias in N predictions.

Liu et al. [136] developed a non-destructive method to monitor the chlorophyll and N content in *Toona sinensis* leaves using NIR and PLS. NIR spectra were paired with reference measurements of chlorophyll and nitrogen content obtained through standard laboratory analyses. The optimal chlorophyll prediction model reached an R^2 of 0.72 with a RMSE of 0.25, indicating reliable predictive power. The N content model, though slightly less accurate, performed well with an R^2 of 0.63 and an RMSE of 0.87, indicating a reasonable balance between explained variance and prediction error.

In conclusion, no definitive studies currently offer rapid alternatives to standard analytical methods for predicting N content in wood chips. However, the methods proposed by the cited authors show potential for adaptation and application in this field. Implementing these approaches could enhance decision-making around optimal harvesting times, improving the sustainability and efficiency of bioenergy operations by providing immediate, on-site data without relying on destructive sampling.

4.4. ML and Heating Value

Developing models for the accurate prediction of HV is essential for the optimization of energy production, cost-minimization and for a sustainable resource management. In order to find a more rapid alternative to traditional laboratory analyses, a number of authors focused on the prediction of HHV.

For instance, Lieskovský et al. [137], presented a foundational study investigating the impact of AC on wood chips' HHV. By establishing a clear relationship between the two parameters through a generalized linear model, the study provided a basis for further modeling efforts that could possibly incorporate additional variables. Results highlighted that for every 1 % increase in AC, HHV decreased by approximately 0.1 MJ/kg. However, the statistical analysis revealed that AC accounted for only 49 % of the variability in HHV, suggesting that other important factors influencing energy content were not captured by the model.

Nieto et al. [138] developed a predictive model combining SVM with particle swarm optimization (PSO) to estimate the higher heating value (HHV) of biomass during torrefaction, a pretreatment that enhances biomass fuel quality. By using key properties as input variables, such as fixed carbon, volatile matter, reaction temperature, residence time, and the atomic O/C and H/C ratios, the study highlighted the effectiveness of this PSO-optimized SVM model, which achieved a high R^2 of 0.94. When compared with ANN and RF models, the PSO-SVM model performed best, offering an efficient and reliable tool for enhancing biomass energy production processes.

Noushabadi et al. [139], applied multiple regression models, such as ANN, SVR and NR, to predict HHV of biomasses via C, H, N, O and S data. The NR model achieved an R^2 of 0.95 in testing, with an RMSE of 1.04 MJ/kg and a standard deviation of 3.58 across the total dataset, reflecting strong accuracy and consistency. The SVR model had an R^2 of 0.96 in testing, with an RMSE of 1.10 MJ/kg and a standard deviation of 3.65, showing slightly more variation in predictions. The ANN model also performed well, with an R^2 of 0.95 in testing, an RMSE of 1.08 MJ/kg, and a standard deviation of 3.76, indicating high accuracy but a slightly broader spread in predictions compared to MNR. Overall, MNR and ANN showed comparable performance, with MNR offering a slight edge in consistency across the total dataset. It should also be noted that models built from literature data may have limitations due to inconsistencies and the potential use of outdated information.

NIR spectroscopy combined with PLS regression is a widely used method in the literature for predicting HHV, often applied alongside the prediction of other key parameters. This approach leverages the strengths of NIR's rapid, non-destructive analysis and the robust predictive capabilities of PLS regression, making it a preferred choice in biomass studies and similar applications.

For instance, Saha et al. [140] developed a PLS model for the prediction of mixed pine wood species, specifically *Pinus taeda* and *Pinus*

elliottii, utilizing NIR spectra as predictor variables alongside laboratory data for reference. Upon testing the model with validation data, it demonstrated robust performance, achieving an R^2 value of 0.89 and a bias of -7.35 Cal/g. These results indicate a strong correlation between the HHV values predicted by the NIR spectra and the actual laboratory measurements [141]. However, the study's focus on a specific wood type limits the applicability of its findings to broader contexts.

Lastly, in a previously discussed study by Mancini et al. [130], HHV data of 125 wood chip samples were coupled with NIR spectroscopy and PLS techniques in order to model HHV. Various pre-processing techniques were evaluated to improve the model's accuracy. The optimal cross validated model achieved a high predictive performance with an R^2 value of 0.81 and a RMSE of 289 J/g. This indicates that the model can predict HHV with a standard error of 234 J/g, offering a rapid and cost-effective alternative to traditional laboratory methods.

In conclusion, various modeling approaches, including ANN, SVM, and NIR spectroscopy combined with PLS, have shown differing levels of success in predicting HHV. However, each method has its limitations. The accuracy of these models is closely tied to the quality of the input data and the diversity of the samples used. This underscores the necessity for further research to develop more robust and generalizable models that integrate multiple parameters.

5. Application of Machine Learning in the wood chip production chain

This section investigates the potential of utilizing ML models to monitor woodchip quality throughout the production chain.

The wood energy production chain starts in the forest. Typically, wood chips are produced by on-site chipping of by-products of silvicultural activities or from dedicated plantations [142]. Before felling, various innovative methods can be employed in forest management to enhance the quality and sustainability of chip production.

For instance, using on-site destructive analyses, Güner et al. [143] developed NR and ANN models to successfully predict the yield of black pine (*Pinus nigra*) forests, using the weights of the various components of the tree served as dependent variables, and tree height and diameter as independent variables. Despite their species-specific nature, these models provide valuable insights for predicting the mass and energy yield of forests. Combined with site-specific data, such as tree density per hectare, they could enable the selection of the optimal felling period. Furthermore, to facilitate rapid and timely productivity predictions, site-specific data can be gathered through satellite imagery and unmanned aerial vehicles [144,145].

After felling, forest by-products are typically chipped on-site, often at the location of the felling [146,147]. At this stage, various technologies and models can be employed to monitor chip quality in real time during the chipping process. For example, the radio frequency method, combined with ML as discussed in the referenced study by Daassi-Gnaba et al. [114], could be further refined for in-situ application on freshly produced wood chip piles. This approach could facilitate the categorization of biomass into MC classes, offering potential benefits not only for organizing chip piles but also for optimizing chip loads, managing chipper feedstock, and improving planning for fuel deliveries, storage, and combustion efficiency.

Radio frequency-based systems could be backed by the widely explored NIR systems. A prominent example is the in-line NIR system developed by Nascimbem et al. [148], which integrated NIR spectroscopy with PLS-DA to assess the MC of wood chips. Once classified by MC, SVC has been utilized to further evaluate the quality of the wood chips within each MC class, resulting in a more nuanced and accurate characterization of their properties.

A further parameter that could be monitored directly post-chipping is the particle size distribution. Wood chips' PSD has an impact on many aspects, ranging from the transportability [149] to the degradation during storage [150]. In fact, PSD influences the arrangement of the

individual chip flakes when the chips are placed in the pile, which has a direct effect on the air flow in the pile, thus on the degradation processes in the pile [150,151]. In this context, Hartmann et al. [42] used a camera-based system and image processing techniques to automatically assess the PSD of wood chips in dynamic movements. Monitoring PSD is crucial as it should be checked as close to the chipping operation as possible. This allows for immediate adjustments, such as changing blades or tuning the equipment, as soon as the particle sizes deviate from specifications, helping to prevent financial losses.

All the above-mentioned studies could be implemented directly during the chipping stage, having an imaging system to monitor the PSD of the chips directly outside the chipper's ejection port, and a NIR or radio frequency system for an initial qualitative assessment of the pile formed directly by the chipper.

Finally, during the storage phase, whether in the forest following chipping or directly at the processing plant, the implementation of new technologies and ML could play a crucial role in monitoring the biological stability of the wood chip piles [152]. In this context, previously discussed methods for the prediction of MC content or classification into MC classes could be used [114,117,118]. Lastly, research by Mokroš et al. [6], which utilized UAVs for estimating the volume of wood chip piles, demonstrates the potential of UAVs equipped with sensors [153] for rapid MC assessment of stocked piles.

Integrating ML models with innovative analytical techniques represents a game-changing approach to quality control in woodchip production. Real-time, continuous assessment methods can significantly reduce both the costs and time typically required for laboratory testing, while also improving the precision of quality evaluations and streamlining storage and processing operations.

As the industry faces stricter emission regulations, these technological advancements become even more critical. Biomass combustion, while sustainable, can release harmful pollutants into the atmosphere. To address this, producers must adopt strategies that not only comply with regulations but also prevent emissions at the source. By improving predictive accuracy and enhancing control algorithms, it becomes possible to minimize pollutant release during combustion, optimizing performance and sustainability.

This shift toward advanced predictive capabilities and emission prevention highlights the broader impact of these innovations, which promise to support both economic efficiency and environmental responsibility in the biomass energy sector.

6. Conclusion

The implementation of ML in wood chip quality monitoring is becoming increasingly viable. While current studies offer models for predicting or classifying wood chip quality, many of these models rely on limited datasets. Collecting data from standard measurements, when combined with alternative data sources such as spectral responses, is crucial for developing robust models. However, obtaining standard measurements is often costly and time-consuming. Therefore, future research should explore novel techniques not yet employed in the existing literature to address the challenges posed by sparse datasets.

One promising solution is data augmentation, which can enhance ML model training by artificially increasing the size and diversity of the dataset. Methods such as synthetic data generation, where new data points are created based on existing data, or adding noise to current measurements, can simulate a broader range of samples and help models generalize more effectively.

In addition, semi-supervised learning techniques can be employed to reduce the reliance on fully labeled datasets. Semi-supervised learning utilizes both labeled and unlabeled data to strengthen the learning process. By incorporating large amounts of unlabeled data, these techniques can improve model accuracy and generalization with fewer labeled samples. This is especially valuable in wood chip quality monitoring, where obtaining labeled data may be difficult or costly.

Furthermore, ML offers many advantages over traditional statistical and mathematical methods for improving fuel handling and wood chip quality. Unlike traditional approaches that rely on fixed assumptions and often miss complex relationships, ML can uncover hidden patterns in data and adapt to changes, such as seasonal variations in biomass quality. Using data from sensors that measure moisture, particle size, and density, ML can help ensure consistent chip quality, more accurately predict energy output, and make real-time adjustments to handling processes. The result is improved efficiency, reduced waste, and more reliable operations.

Overcoming the challenges of limited and costly datasets will not only improve model performance but also make the technology more accessible and scalable, ultimately supporting more sustainable and cost-effective practices in the woody biofuel industries.

CRedit authorship contribution statement

Thomas Gasperini: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Conceptualization. **Volkan Yeşil:** Writing – review & editing. **Giuseppe Toscano:** Validation, Supervision, Conceptualization.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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